

Euler's Method

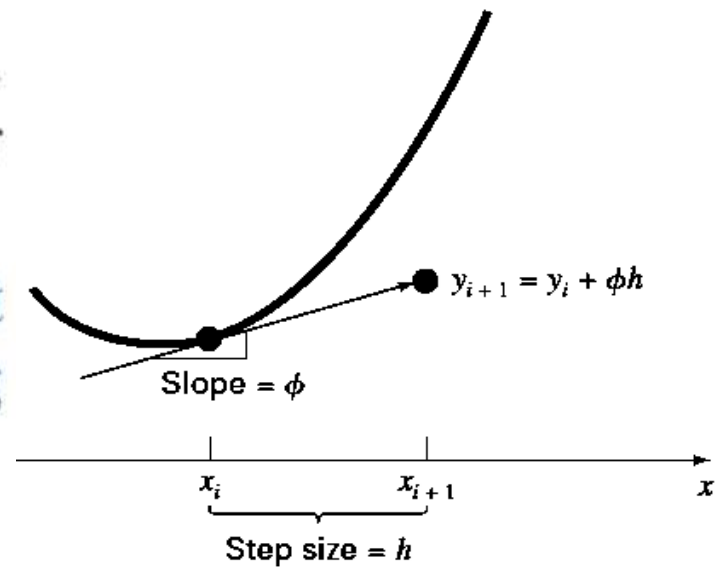
Euler's method is the most elementary approximation technique for solving initial-value problems. Although it is seldom used in practice, the simplicity of its derivation can be used to illustrate the techniques involved in the construction of some of the more advanced techniques, without the cumbersome algebra that accompanies these constructions.

The object of Euler's method is to obtain approximations to the well-posed initial-value problem

$$\frac{dy}{dt} = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha.$$

A continuous approximation to the solution $y(t)$ will not be obtained; instead, approximations to y will be generated at various values, called **mesh points**, in the interval $[a, b]$. Once the approximate solution is obtained at the points, the approximate solution at other points in the interval can be found by interpolation.

We first make the stipulation that the mesh points are equally distributed throughout the interval $[a, b]$. This condition is ensured by choosing a positive integer N and selecting the mesh points



$$\frac{dy}{dt} = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha.$$

We will use Taylor's Theorem to derive Euler's method. Suppose that $y(t)$, the unique solution

$$y(t_{i+1}) = y(t_i) + (t_{i+1} - t_i)y'(t_i) + \frac{(t_{i+1} - t_i)^2}{2}y''(\xi_i),$$

for some number ξ_i in (t_i, t_{i+1}) . Because $h = t_{i+1} - t_i$, we have

$$y(t_{i+1}) = y(t_i) + hy'(t_i) + \frac{h^2}{2}y''(\xi_i),$$

and, because $y(t)$ satisfies the differential equation

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + \frac{h^2}{2}y''(\xi_i).$$

Euler's method constructs $w_i \approx y(t_i)$, for each $i = 1, 2, \dots, N$, by deleting the remainder term. Thus Euler's method is

$$\begin{aligned} w_0 &= \alpha, \\ w_{i+1} &= w_i + hf(t_i, w_i), \quad \text{for each } i = 0, 1, \dots, N-1. \end{aligned}$$

difference equation associated with Euler's method.

Initial Value Problem

$$y' = f(t, y) \quad a \leq t \leq b,$$

$$y(a) = \alpha$$

Euler's Method

$$y(t_{i+1}) \approx y(t_i) + hf(t_i, y(t_i))$$

$$y(t_0) = \alpha$$

Euler's Method

$$w_{i+1} = w_i + hf(t_i, w_i)$$

$$w_0 = \alpha$$

$$y(t_i) \approx w_i$$

$$y(t_{i+1}) \approx w_{i+1}$$

Example

use an algorithm for Euler's method to approximate the solution to

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5,$$

at $t = 2$. Here we will simply illustrate the steps in the technique when we have $h = 0.5$.

For this problem $f(t, y) = y - t^2 + 1$, so

$$w_0 = y(0) = 0.5;$$

$$w_1 = w_0 + 0.5(w_0 - (0.0)^2 + 1) = 0.5 + 0.5(1.5) = 1.25;$$

$$w_2 = w_1 + 0.5(w_1 - (0.5)^2 + 1) = 1.25 + 0.5(2.0) = 2.25;$$

$$w_3 = w_2 + 0.5(w_2 - (1.0)^2 + 1) = 2.25 + 0.5(2.25) = 3.375;$$

$$y(2) \approx w_4 = w_3 + 0.5(w_3 - (1.5)^2 + 1) = 3.375 + 0.5(2.125) = 4.4375.$$

Example

Solve the following initial value problem over the interval from $t = 0$ to 2 where $y(0) = 1$.

(a) Euler's method with $h = 0.5$ and 0.25 .

$$\frac{dy}{dt} = yt^3 - 1.5y$$

0.0000	1.0000
0.5000	0.2500
1.0000	0.0781
1.5000	0.0586
2.0000	0.1135

$$w_{i+1} = w_i + h(w_i t_i^3 - 1.5w_i) \quad \text{Euler's Method}$$

$$w_0 = 1$$

$$w_1 = w_0 + 0.5(w_0(0^3) - 1.5w_0)$$

$$= 1 + 0.5(0 - 1.5(1.0)) = 0.25$$

$$w_2 = w_1 + 0.5(w_1(0.5)^3 - 1.5w_1)$$

$$= 0.25 + 0.5(0.25 * 0.5^3 - 1.5 * 0.25) = 0.0781$$

$$y(0.0) = 1.0$$

$$y(0.5) \approx 0.25$$

$$y(1.0) \approx 0.0781$$

To interpret Euler's method geometrically, note that when w_i is a close approximation to $y(t_i)$, the assumption that the problem is well-posed implies that

$$f(t_i, w_i) \approx y'(t_i) = f(t_i, y(t_i)).$$

The graph of the function highlighting $y(t_i)$ is shown in Figure 5.2. One step in Euler's method appears in Figure 5.3, and a series of steps appears in Figure 5.4.

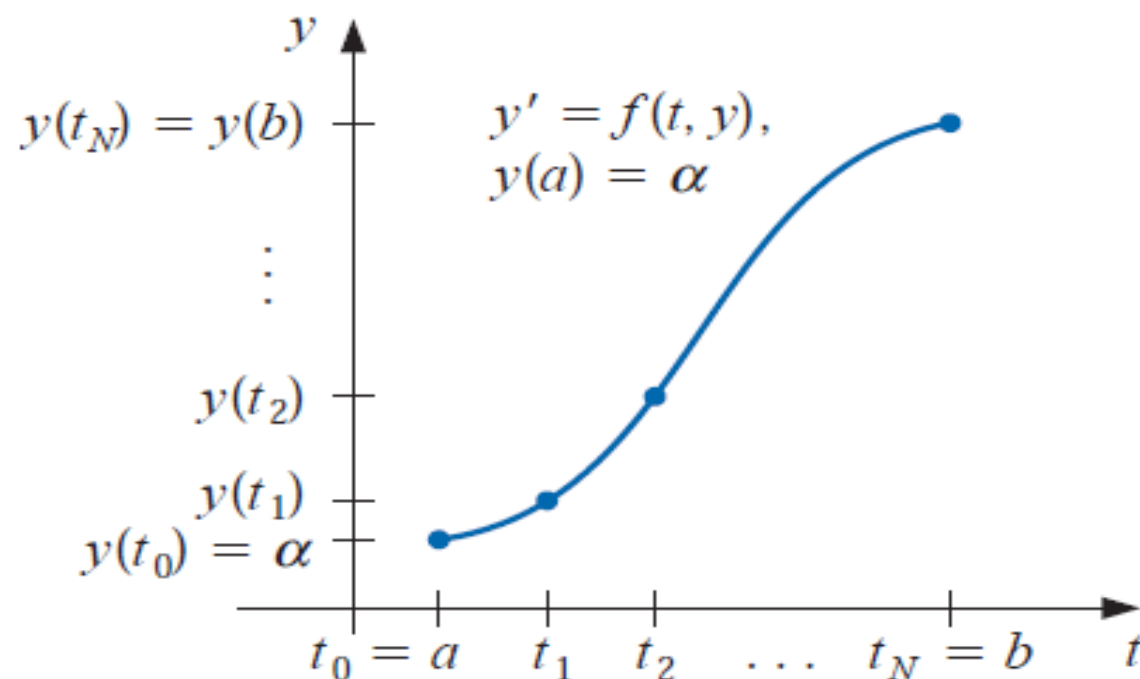


Figure 5.3

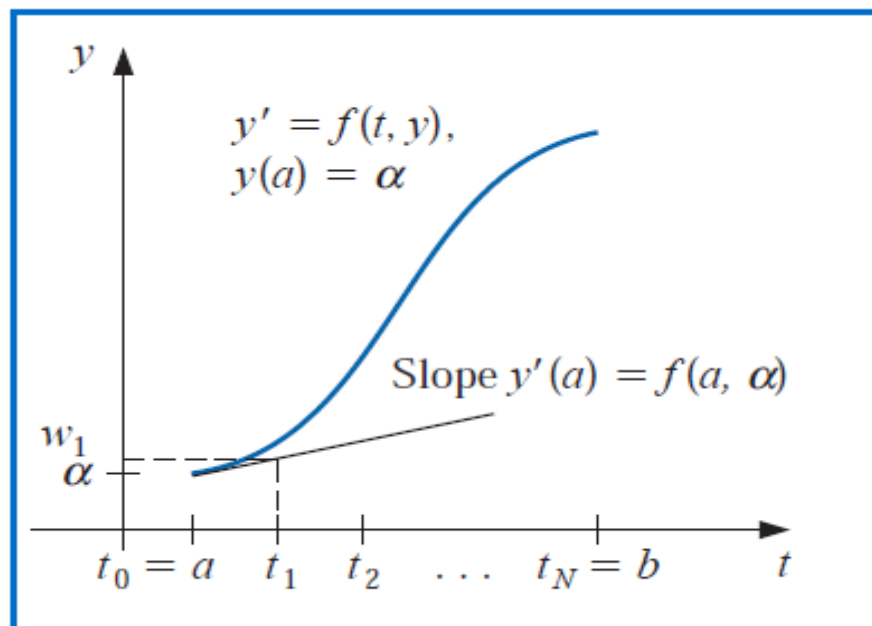
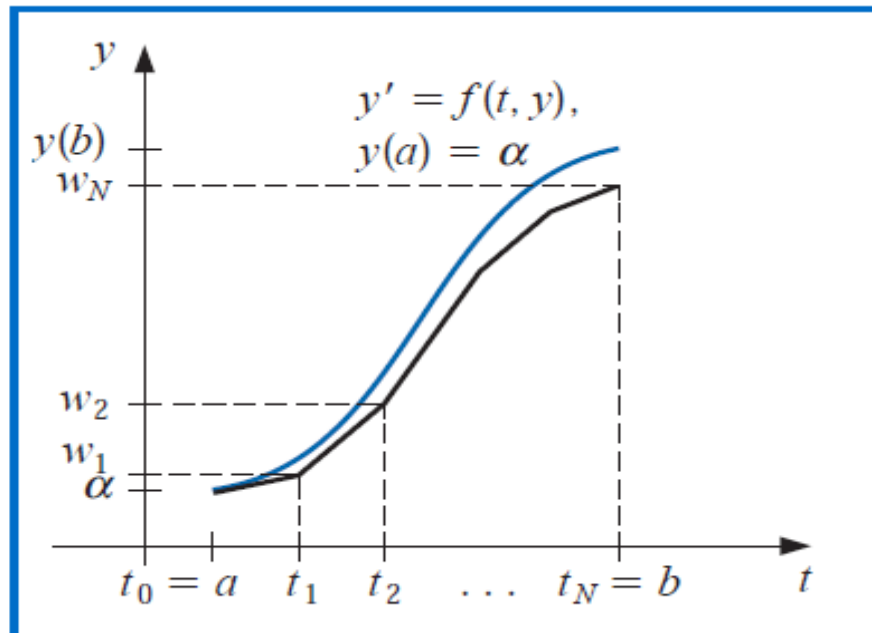


Figure 5.4



Theorem 5.9 Suppose f is continuous and satisfies a Lipschitz condition with constant L on

Error Bounds for Euler's Method

$$D = \{(t, y) \mid a \leq t \leq b \text{ and } -\infty < y < \infty\}$$

and that a constant M exists with

$$|y''(t)| \leq M, \quad \text{for all } t \in [a, b],$$

where $y(t)$ denotes the unique solution to the initial-value problem

$$y' = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha.$$

Let $w_0, w_1, \dots, w_N$ be the approximations generated by Euler's method for some positive integer N . Then, for each $i = 0, 1, 2, \dots, N$,

$$|y(t_i) - w_i| \leq \frac{hM}{2L} [e^{L(t_i-a)} - 1].$$

Example 2 The solution to the initial-value problem

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5,$$

was approximated in Example 1 using Euler's method with $h = 0.2$. Use the inequality in Theorem 5.9 to find a bounds for the approximation errors and compare these to the actual errors.

Solution Because $f(t, y) = y - t^2 + 1$, we have $\partial f(t, y)/\partial y = 1$ for all y , so $L = 1$. For this problem, the exact solution is $y(t) = (t + 1)^2 - 0.5e^t$, so $y''(t) = 2 - 0.5e^t$ and

$$|y''(t)| \leq 0.5e^2 - 2, \quad \text{for all } t \in [0, 2].$$

Using the inequality in the error bound for Euler's method with $h = 0.2$, $L = 1$, and $M = 0.5e^2 - 2$ gives

$$|y_i - w_i| \leq 0.1(0.5e^2 - 2)(e^{t_i} - 1).$$

Hence

$$|y(0.2) - w_1| \leq 0.1(0.5e^2 - 2)(e^{0.2} - 1) = 0.03752;$$

$$|y(0.4) - w_2| \leq 0.1(0.5e^2 - 2)(e^{0.4} - 1) = 0.08334;$$

t_i	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
Actual Error	0.02930	0.06209	0.09854	0.13875	0.18268	0.23013	0.28063	0.33336	0.38702	0.43969
Error Bound	0.03752	0.08334	0.13931	0.20767	0.29117	0.39315	0.51771	0.66985	0.85568	1.08264

Figure 5.3

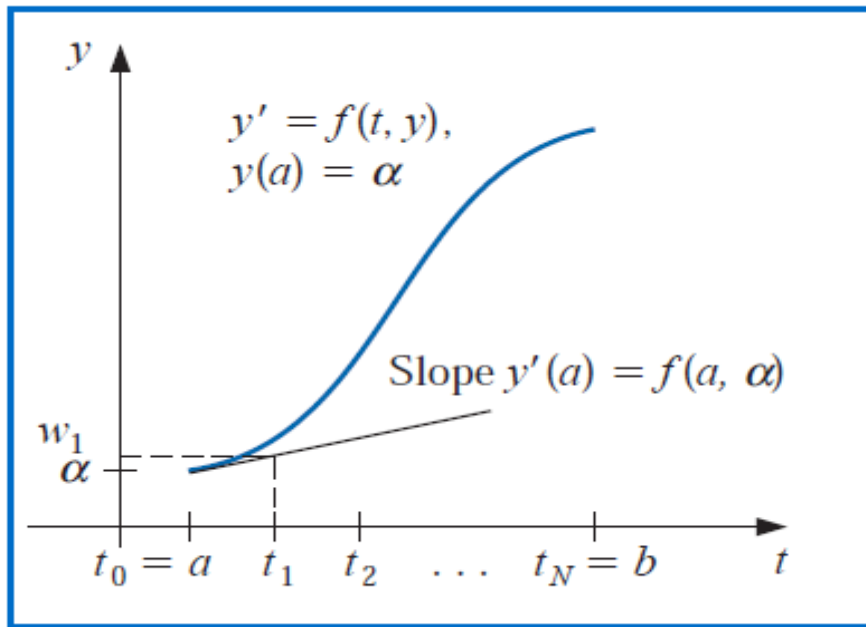
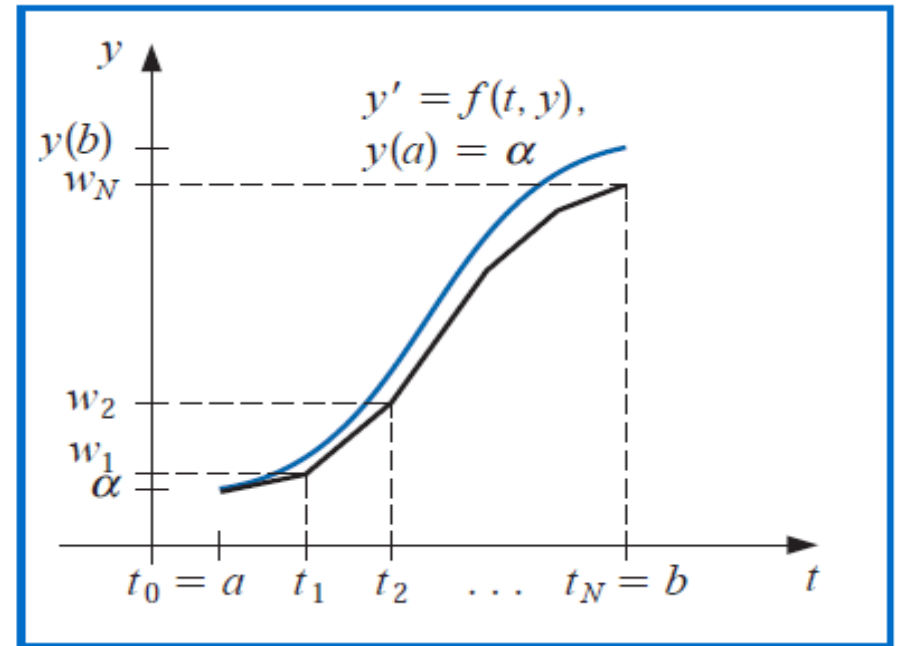


Figure 5.4



We will see modifications of Euler's Method. All these techniques are generally called *Runge-Kutta* methods.

Another simple modification of Euler's method: the *midpoint method*

This technique uses Euler's method to predict a value of y at the midpoint. The midpoint method is superior to Euler's method because it utilizes a slope estimate at the midpoint of the prediction interval. Although there are certain cases where easily programmable techniques such as Euler's method can be applied to advantage, the Heun and midpoint methods are generally superior and should be implemented if they are consistent with the problem objectives. The Heun, the midpoint method, and in fact, the Euler technique itself are versions of a broader class of one-step approaches called Runge-Kutta methods.

RUNGE-KUTTA METHODS

Modified Euler Method

$$w_{i+1} = w_i + \frac{h}{2} (K_1 + K_2)$$

$$w_0 = \alpha$$

$$K_1 = f(t_i, w_i)$$

$$K_2 = f(t_i + h, w_i + K_1 h)$$

Midpoint method

$$w_{i+1} = w_i + hK_2$$

$$w_0 = \alpha$$

$$K_1 = f(t_i, w_i)$$

$$K_2 = f\left(t_i + \frac{h}{2}, w_i + \frac{h}{2} K_1\right)$$

Euler's Method

$$w_{i+1} = w_i + hK_1$$

$$w_0 = \alpha$$

$$K_1 = f(t_i, w_i)$$

Heun's Method

$$w_{i+1} = w_i + \frac{h}{4} (K_1 + 3K_3)$$

$$w_0 = \alpha$$

$$K_1 = f(t_i, w_i)$$

$$K_2 = f\left(t_i + \frac{h}{3}, w_i + \frac{h}{3} K_1\right)$$

$$K_3 = f\left(t_i + \frac{2h}{3}, w_i + \frac{2h}{3} K_2\right)$$

RUNGE-KUTTA METHODS

Example

Use the Modified Euler method
 $h = 0.5$, to approximate $y(1)$

$$\begin{aligned}y' &= y - t^2 + 1 \\ 0 &\leq t \leq 4, \\ y(0) &= 0.5\end{aligned}$$

Modified Euler Method

$$\begin{aligned}w_{i+1} &= w_i + \frac{h}{2} (K_1 + K_2) \\ w_0 &= \alpha\end{aligned}$$

$$K_1 = f(t_i, w_i)$$

$$K_2 = f(t_i + h, w_i + K_1 h)$$

$$f(t, y) = y - t^2 + 1$$

$$\begin{aligned}K_1 &= f(t_0, w_0) \\ &= f(0, 0.5) = 1.5\end{aligned}$$

$$\begin{aligned}K_2 &= f(t_0 + h, w_0 + K_1 h) \\ &= f(0.5, 0.5 + 1.5 * 0.5) = 2\end{aligned}$$

$$w_1 = 0.5 + \left(\frac{1}{2} * 1.5 + \frac{1}{2} * 2 \right) 0.5$$

$$w_1 = 1.375 \rightarrow y(0.5) \approx 1.375$$

$$K_1 = f(t_1, w_1)$$

$$= f(0.5, 1.375) = 2.125$$

$$K_2 = f(t_1 + h, w_1 + K_1 h)$$

$$\begin{aligned}&= f(1, 1.375 + 2.125 * 0.5) \\ &= 2.4375\end{aligned}$$

$$w_2 = 1.375 + \left(\frac{1}{2} * 2.125 + \frac{1}{2} * 2.4375 \right) 0.5$$

$$w_2 = 2.515625 \rightarrow y(1) \approx 2.515625$$

RUNGE-KUTTA METHODS

Example

Use the Midpoint method $h = 0.5$,
to approximate $y(0.5)$

$$\begin{aligned}y' &= y - t^2 + 1 \\ 0 \leq t &\leq 4, \\ y(0) &= 0.5\end{aligned}$$

Midpoint method

$$\begin{aligned}w_{i+1} &= w_i + hK_2 \\ w_0 &= \alpha\end{aligned}$$

$$K_1 = f(t_i, w_i)$$

$$K_2 = f\left(t_i + \frac{h}{2}, w_i + \frac{h}{2}K_1\right)$$

Midpoint method

$$\begin{aligned}w_{i+1} &= w_i + hf\left(t_i + \frac{h}{2}, w_i + \frac{h}{2}f(t_i, w_i)\right) \\ w_0 &= \alpha\end{aligned}$$

$$f(t, y) = y - t^2 + 1$$

$$\begin{aligned}K_1 &= f(t_0, w_0) \\ &= f(0, 0.5) = 1.5\end{aligned}$$

$$K_2 = f(t_0 + 0.5 * h, w_0 + 0.5 * K_1 h)$$

$$= f(0.5, 0.5 + 0.5 * 1.5 * 0.5) = 1.8125$$

$$w_1 = 0.5 + (1.8125)0.5 = 1.40625$$

$$\rightarrow y(0.5) \approx 1.40625$$

RUNGE-KUTTA METHODS

Example

Use the Heun's method with $h = 0.2$, to approximate $y(0.2)$

$$\begin{aligned}y' &= y - t^2 + 1 \\ 0 &\leq t \leq 2, \\ y(0) &= 0.5\end{aligned}$$

Heun's Method

$$w_{i+1} = w_i + \frac{h}{4} (\mathbf{K}_1 + 3\mathbf{K}_3)$$

$$w_0 = \alpha$$

$$\mathbf{K}_1 = f(t_i, w_i)$$

$$\mathbf{K}_2 = f\left(t_i + \frac{h}{3}, w_i + \frac{h}{3}\mathbf{K}_1\right)$$

$$\mathbf{K}_3 = f\left(t_i + \frac{2h}{3}, w_i + \frac{2h}{3}\mathbf{K}_2\right)$$

$$f(t, y) = y - t^2 + 1$$

$$\begin{aligned}\mathbf{K}_1 &= f(t_0, w_0) \\ &= f(0, 0.5) = 1.5\end{aligned}$$

$$\begin{aligned}\mathbf{K}_2 &= f(t_0 + 1/15, w_0 + 1.5/15) \\ &= 1.5956\end{aligned}$$

$$\begin{aligned}\mathbf{K}_3 &= f(t_0 + 2/15, w_0 + (2/15)(1.5956)) \\ &= 1.694969\end{aligned}$$

$$\begin{aligned}w_1 &= 0.5 + \frac{0.2}{4} (\mathbf{1.5} + 3 * \mathbf{1.694969}) \\ w_1 &= 0.82924 \rightarrow \mathbf{y(0.2) \approx 0.82924}\end{aligned}$$

The most popular method is the fourth order Runge-Kutta method, or RK4 method

Fourth-Order Runge-Kutta Methods

$$W_{i+1} = W_i + \frac{1}{6} (\textcolor{blue}{K}_1 + 2\textcolor{red}{K}_2 + 2\textcolor{green}{K}_3 + \textcolor{brown}{K}_4)$$

$$W_0 = \alpha$$

$$\textcolor{blue}{K}_1 = h F(t_i, W_i)$$

$$\textcolor{red}{K}_2 = h F\left(t_i + \frac{h}{2}, W_i + \frac{1}{2}\textcolor{blue}{K}_1\right)$$

$$\textcolor{green}{K}_3 = h F\left(t_i + \frac{h}{2}, W_i + \frac{1}{2}\textcolor{red}{K}_2\right)$$

$$\textcolor{brown}{K}_4 = h F(t_{i+1}, W_i + \textcolor{green}{K}_3)$$

Runge-Kutta Methods

$O(h)$ 1) Euler Method

$O(h^2)$ 2) Modified Euler Method

$O(h^2)$ 3) Midpoint Method

$O(h^3)$ 4) Heun's Method

$O(h^4)$ 5) RK4 Method

RUNGE-KUTTA METHODS

Example

Use the RK4 method with $h = 0.2$, to approximate $y(0.2)$

$$\begin{aligned}y' &= y - t^2 + 1 \\ 0 &\leq t \leq 2, \\ y(0) &= 0.5\end{aligned}$$

Fourth-Order Runge-Kutta Methods

$$W_{i+1} = W_i + \frac{1}{6}(\mathbf{K}_1 + 2\mathbf{K}_2 + 2\mathbf{K}_3 + \mathbf{K}_4)$$

$$W_0 = \alpha$$

$$\mathbf{K}_1 = h F(t_i, W_i)$$

$$\mathbf{K}_2 = h F\left(t_i + \frac{h}{2}, W_i + \frac{1}{2}\mathbf{K}_1\right)$$

$$\mathbf{K}_3 = h F\left(t_i + \frac{h}{2}, W_i + \frac{1}{2}\mathbf{K}_2\right)$$

$$\mathbf{K}_4 = h F(t_{i+1}, W_i + \mathbf{K}_3)$$

$$w_0 = 0.5$$

$$k_1 = 0.2f(0, 0.5) = 0.2(1.5) = 0.3$$

$$k_2 = 0.2f(0.1, 0.65) = 0.328$$

$$k_3 = 0.2f(0.1, 0.664) = 0.3308$$

$$k_4 = 0.2f(0.2, 0.8308) = 0.35816$$

$$\begin{aligned}w_1 &= 0.5 + \frac{1}{6}(0.3 + 2(0.328) + 2(0.3308) + 0.35816) \\ &= 0.8292933.\end{aligned}$$